Scalable Probit Share Calibration

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Abstract

Probit models are a cornerstone of economic, choice, and contest theory, and machine learning runs on the same latent-utility family. Yet the many-alternative form sees little use beyond small menus: its probabilities have no closed form, the standard simulator prices one alternative at a time, and calibration, meaning the recovery of the utilities at which model shares match observed shares with the factor covariance given and fixed, has been out of reach at scale. We give an algorithm that calibrates a two-factor probit model with a thousand alternatives in three seconds on a laptop, with cost growing linearly in the number of alternatives: roughly a million times faster than calibrating through the best simulation alternative at equivalent accuracy. The computed shares carry log-odds errors of a few parts in a hundred thousand, hundred-to-one tails included.

Keywords: multinomial probit; discrete choice; numerical quadrature; Jacobian–vector product; share inversion; factor model

1 Introduction

Probit models are a cornerstone of economic, choice, and contest theory: Thurstone's comparative judgment, the ordered and binary probits of applied microeconometrics, and the Lazear–Rosen tournament are all the same Gaussian latent-performance model. Machine learning runs on the same family: a softmax output layer is its Luce–Gumbel member, and Thurstone–Mosteller pairwise comparison, the probit member, underlies rating systems and preference learning.

Such models are calibrated to observed shares so that the recovered utilities can answer questions the data do not: elasticities, welfare, pricing, the effect of adding or removing an alternative. The models used for this, essentially universally, are the logit family: multinomial logit, nested logit, and above all random-coefficients logit in the BLP tradition (Berry et al., 1995; Train, 2009; Conlon and Gortmaker, 2020). They are universal for one reason. They can be computed.

*Every number in this paper is produced by a committed, seeded script; see the Reproducibility section and `https://github.com/microprediction/kinetics`.

Choice modeling did not start there: it started with Thurstone's 1927 model (Thurstone, 1927) of jointly Gaussian utilities with unrestricted correlation, known in econometrics as multinomial probit (MNP).

Probit probabilities are $(N-1)$-dimensional Gaussian integrals with no closed form. The standard evaluator is the Geweke–Hajivassiliou–Keane (GHK) simulator (Geweke et al., 1994; Hajivassiliou and McFadden, 1998; Train, 2009), the econometric branch of the Genz separation-of-variables family (Genz, 1992) whose RQMC form (Genz–Bretz) is the default in computational statistics: it writes the probability that alternative i beats the rest as an orthant probability of the difference utilities, and estimates it by sampling truncated normals sequentially along a Cholesky factor. That is a separate simulation per alternative, with $R^{-1/2}$ error in the number of draws, and it is why probit survives mainly on small menus: the field kept the computable models and gave up the model it started from.

The concession is not free. Logit-family substitution is structurally restricted, and the calibration apparatus that demand estimation is built on is routine only on the logit side. One measure of the cost, from one deliberately misspecified synthetic design with oracle factor loadings (Section 4): in the largest-deletion stratum, plain logit misallocates about 23% of the demand released by a deleted alternative, against 5% for factor probit, the closest of the four candidate structures.

This paper's point is that the barrier is not real for an important class of covariance structures, the low-rank-plus-diagonal factor model $\Sigma = VV^\top + \mathrm{diag}(D)$. The goal is **calibration**: given observed shares and a fixed, known factor structure (V, D), recover the utilities at which the probit model reproduces them.

This is the probit analogue of the inversions demand estimation runs on the logit side every day (Berry et al., 1995; Conlon and Gortmaker, 2020); estimating (V, D) is the outer problem, discussed in Section 6.

We solve it at $N = 1000$ in three seconds, and the inverse problem is globally well-posed (Proposition 4: shares determine utilities uniquely up to location). Three ingredients make it work:

1. **A fast forward map** (utilities $\to$ shares): all N probabilities at once in $O(QNL)$ for Q factor nodes and L lattice points. Calibration needs many forward evaluations; this is what makes them affordable.

2. **Resampling-free derivatives**: Jacobian–vector products in $O(QNL)$; in max-wins coordinates the Jacobian is a weighted graph Laplacian, so reduced systems are symmetric positive definite. This supports implicit differentiation and Newton–Krylov solvers; the calibration reported below uses a cheaper damped Jacobi iteration.

3. **A single shared construction**: conditionally on the k factors the alternatives are independent, and a lattice survival-product identity computes every alternative's integral from one *shared survival field*, built once and reused for shares, slopes, and all N single-removal counterfactuals.

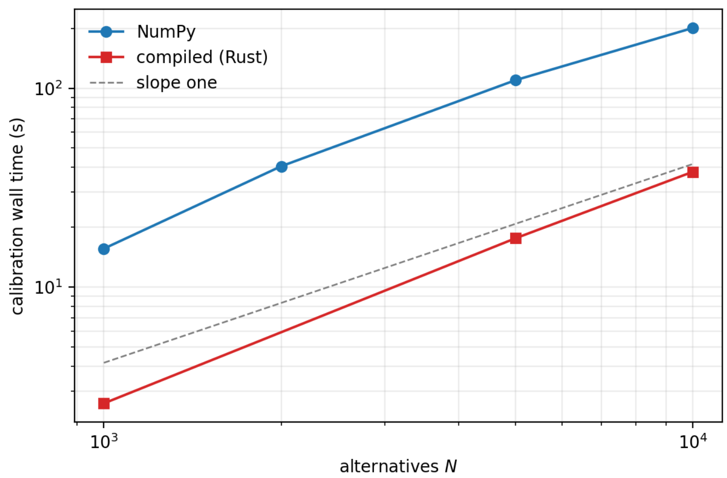

Figure 1: Wall time to calibrate the full utility vector from target shares ($k = 2$ factors, independent 5×10^6-draw targets, experiment 19). Both implementations grow roughly linearly in N; the dashed guide has slope one.

Prior art. The nearest methods, and what each lacks:

approach	all N shares	calibration
GHK simulation (Geweke et al., 1994; Hajivassiliou and McFadden, 1998)	N separate runs, $R^{-1/2}$ error	not attempt
direct utility simulation	whole vector per draw, $R^{-1/2}$	no derivativ
analytic approximations (Bhat, 2011; Connors et al., 2014)	per-alternative	no
factor quadrature (Butler and Moffitt, 1982; Elrod and Keane, 1995)	N separate $(k{+}1)$-dim integrals	no
Bayesian factor MNP (Loaiza-Maya and Nibbering, 2022)	simulation inside estimation	posterior, no
neural amortization (Huch and Keane, 2026)	fast, uncontrolled error	no
BLP-side inversion (Berry et al., 1995; Conlon and Gortmaker, 2020)	—	logit family
this paper	one $O(QNL)$ pass	Prop. 4; 3 s

Figure 1 shows calibration wall time as the number of alternatives grows: roughly linear, three seconds at a thousand alternatives and forty at ten thousand in the compiled implementation.

Scope. The method needs the factor form: where Σ has no adequate low-rank-plus-diagonal approximation in the choice-relevant quotient space, simulation at exact Σ remains the tool (Section 4). The decisive advantages are precision, reproducibility, resampling-free derivatives, and the calibration machinery.

The one-pass identity comes from an unlikely literature: inferring ability from win probabilities in multi-entrant races (Cotton, 2021). That is perhaps why it has not been assembled for probit before, despite each ingredient being classical.

Model. Throughout,

$$U_i \;=\; \mu_i + v_i^\top f + \sqrt{D_i}\,\varepsilon_i, \qquad f \sim N(0, I_k), \quad \varepsilon_i \sim N(0,1) \text{ i.i.d.}, \quad D_i > 0, \tag{1}$$

i.e. $\Sigma = VV^\top + \mathrm{diag}(D)$; the chosen alternative is the argmax of U. Throughout, (V, D) is supplied and held fixed; calibration solves for μ alone.

Factor dimension reduction for probit is classical: one-factor quadrature goes back to Butler and Moffitt (1982), and factor-analytic probit to Elrod and Keane (1995). The contribution is the all-alternatives assembly, the inverse transform, the derivative structure, and evidence for all of it.

Proposition 1 (Conditional identity)**.** *Under* (1)*, with $g_i(\cdot \mid f)$ and $F_i(\cdot \mid f)$ the conditional density and CDF of U_i given f,*

$$p_i \;=\; \mathbb{E}_f\left[\int g_i(x \mid f) \prod_{j \neq i} F_j(x \mid f)\, dx\right], \qquad \sum_i p_i = 1, \tag{2}$$

and the product over $j \neq i$, for all i simultaneously, is obtained from $\log F_{\text{field}} = \sum_j \log F_j$ *by subtraction.*

Proof. Conditional on f the U_j are independent, so $\mathbb{P}(U_i \text{ maximal} \mid f) = \int g_i \prod_{j \neq i} F_j\, dx$; average over f. The events partition (ties are null for continuous laws), giving $\sum_i p_i = 1$. □

Proposition 2 (Complexity)**.** *With Q factor nodes and L lattice points, the forward share vector costs $O(QN(k+L))$, which is $O(QNL)$ for $k \ll L$; the full single-removal ensemble $\{P(j \text{ chosen} \mid i \text{ removed})\}_{i,j}$ costs $O(QN^2L)$ arithmetic and $O(N^2)$ storage, reusing one conditional field construction.*

Proof. Per node: the conditional locations cost $O(Nk)$, the field log-product $O(NL)$, and each alternative's integral $O(L)$ after subtraction, giving $O(N(k + L))$; the removal ensemble evaluates N survivor integrals for each of N removals, $O(N^2L)$, over the same cached field. □

Proposition 3 (Identification and differentiability)**.** *$p(\mu + c\mathbf{1}) = p(\mu)$, so $J = \partial p/\partial\mu$ has $\mathbf{1}$ in its null space and μ is identified only up to location. With B an $N \times (N-1)$ basis of $\mathbf{1}^\perp$ and $\mu = B\eta$, share inversion and implicit differentiation are taken in reduced coordinates: $d\eta/d\theta = -(B^\top JB)^{-1}B^\top \partial p/\partial\theta$, provided $B^\top JB$ is nonsingular.*

Proof. For the invariance, write $U_i = \mu_i + \xi_i$ with $\xi_i = v_i^\top f + \sqrt{D_i}\,\varepsilon_i$. For every realization of the shocks, $\arg\max_i(\mu_i + c + \xi_i) = \arg\max_i(\mu_i + \xi_i)$, since adding c to every coordinate changes no pairwise comparison. The choice events therefore coincide realization by realization, so $p(\mu + c\mathbf{1}) = p(\mu)$ for all c. Differentiating this identity in c at $c = 0$ gives $J\mathbf{1} = 0$: the ones vector lies in the null space of J, and two utility vectors differing by a common shift produce identical shares, so μ is identified at most up to location.

For the reduced formula, suppose $\eta(\theta)$ satisfies the calibration identity $p(B\eta(\theta), \theta) = \hat{p}$ on an open set of θ. Total differentiation gives the N-equation system

$$JB\,\frac{d\eta}{d\theta} + \frac{\partial p}{\partial\theta} = 0.$$

Only $N-1$ of these equations are independent, but none are lost by projecting: shares sum to one identically in both arguments, so differentiating $\mathbf{1}^\top p = 1$ gives $\mathbf{1}^\top J = 0$ and $\mathbf{1}^\top(\partial p/\partial\theta) = 0$; both terms already lie in $\mathbf{1}^\perp$, so the system is equivalent to its image under $B^\top$. That image is $(B^\top JB)\,d\eta/d\theta = -B^\top\partial p/\partial\theta$, and when $B^\top JB$ is nonsingular (established for this model in Proposition 4) it inverts to the displayed formula, with local existence and smoothness of $\theta \mapsto \eta(\theta)$ supplied by the implicit function theorem. $\square$

Proposition 4 (Shares determine utilities)**.** *For model* (1) *with* $D_i > 0$ *and fixed known* (V, D)*, in max-wins coordinates, define for* $i \neq j$

$$w_{ij} \;=\; \mathbb{E}_f \int g_i(x \mid f)\, g_j(x \mid f) \prod_{\ell \neq i,j} F_\ell(x \mid f)\, dx \;>\; 0.$$

Then $J = \partial p/\partial\mu$ *satisfies* $J_{ij} = -w_{ij}$ *and* $J_{ii} = \sum_{j\neq i} w_{ij}$*, i.e.*

$$J \;=\; \sum_{i<j} w_{ij}(e_i - e_j)(e_i - e_j)^\top, \qquad h^\top Jh \;=\; \sum_{i<j} w_{ij}(h_i - h_j)^2,$$

a weighted graph Laplacian with everywhere-positive weights: J *is symmetric, positive semidefinite, has* $\mathbf{1}$ *as its only null direction, and* $B^\top JB \succ 0$ *for every full-rank basis* B *of* $\mathbf{1}^\perp$*. Moreover* $p = \nabla G$ *for the social-surplus function* $G(\mu) = \mathbb{E}\max_i\{\mu_i + \xi_i\}$ *(the Williams–Daly–Zachary / McFadden identity (McFadden, 1981)), and the restriction of* p *to* $\mathbf{1}^\perp$ *is a smooth bijection onto the interior of the probability simplex: every strictly positive target* $\hat{p}$ *has exactly one mean-zero utility vector.*

Proof. Start from (2). For $j \neq i$, the parameter μ_j enters p_i only through the factor $F_j(x \mid f) = \Phi\big((x - \mu_j - v_j^\top f)/\sqrt{D_j}\big)$, whose μ_j-derivative is $-g_j(x \mid f)$. Differentiating under the integral (dominated convergence: the integrand is bounded by integrable Gaussian tails uniformly in a neighborhood of μ) gives

$$J_{ij} \;=\; \frac{\partial p_i}{\partial \mu_j} \;=\; -\,\mathbb{E}_f \int g_i(x \mid f)\, g_j(x \mid f) \prod_{\ell \neq i,j} F_\ell(x \mid f)\, dx \;=\; -\,w_{ij},$$

and the integrand is symmetric in (i, j), so $J_{ij} = J_{ji}$. For the diagonal, row sums vanish: differentiating $\mathbf{1}^\top p = 1$ in μ_i gives $\sum_j J_{ji} = 0$, hence $J_{ii} = \sum_{j\neq i} w_{ij}$. (The same diagonal follows directly: $\partial_{\mu_i} g_i = -g_i'$, and integrating by parts, $\int(-g_i')\prod_{j\neq i} F_j\, dx = \int g_i \sum_{j\neq i} g_j \prod_{\ell\neq i,j} F_\ell\, dx$, the boundary terms vanishing because $g_i \to 0$ at $\pm\infty$ while the product stays bounded.) Positivity of the weights: for every f and x the Gaussian densities g_i, g_j are strictly positive, and each $F_\ell(x \mid f)$ is strictly positive for finite x, so the integrand is strictly positive on all of $\mathbb{R}$ and $w_{ij} > 0$.

With $J_{ij} = -w_{ij}$ off the diagonal and $J_{ii} = \sum_{j\neq i} w_{ij}$, expanding $\sum_{i<j} w_{ij}(e_i - e_j)(e_i - e_j)^\top$ entry by entry reproduces J, and therefore $h^\top Jh = \sum_{i<j} w_{ij}(h_i - h_j)^2 \geq 0$. Because every w_{ij} is strictly positive, the form vanishes only when $h_i = h_j$ for all pairs, i.e. for constant h: J is positive semidefinite with null space exactly $\operatorname{span}\{\mathbf{1}\}$, hence positive definite on $\mathbf{1}^\perp$. For inversion, minimize $H(\mu) = G(\mu) - \hat{p}^\top\mu$ over $\mathbf{1}^\perp$: H is strictly convex

there (its Hessian is the reduced J), and since the shocks have zero mean, $G(\mu) \geq \max_i \mu_i$, so

$$H(\mu) \;\geq\; \max_i \mu_i - \hat{p}^\top \mu \;=\; \textstyle\sum_i \hat{p}_i(\max_j \mu_j - \mu_i) \;\geq\; \hat{p}_{\min}\,(\max_i \mu_i - \min_i \mu_i) \;\geq\; \hat{p}_{\min}\,\|\mu\|_2/\sqrt{N}$$

on $\mathbf{1}^\perp$, where the last step uses that a zero-sum vector has $\max_i \mu_i \geq 0 \geq \min_i \mu_i$, so the range dominates $\|\mu\|_\infty \geq \|\mu\|_2/\sqrt{N}$; hence H is coercive. Since $D_i > 0$ the utilities have a smooth full-support joint density and ties are null, so differentiating under the expectation gives $\nabla G = p$. At the unique constrained minimizer, $B^\top(p(\mu) - \hat{p}) = 0$, so $p(\mu) - \hat{p} \in \operatorname{span}\{\mathbf{1}\}$; both vectors sum to one, so the coefficient is zero and $p(\mu) = \hat{p}$. Smoothness of the inverse follows from the inverse-function theorem, since $B^\top J B \succ 0$ everywhere. □

Remark 1 (The photo-finish graph). In English: w_{ij} is the probability density of a photo-finish between i and j, a tie for the win with everyone else behind, and moving utility from j to i moves share along every photo-finish edge at exactly that rate. The alternatives form a *photo-finish graph*: complete, with strictly positive weights, because Gaussian utilities give every pair some chance of a dead heat. The proposition's conclusions are then graph facts. Constants span the null space because the quadratic form $\sum_{i<j} w_{ij}(h_i - h_j)^2$ sees only differences: shares cannot identify the overall utility level, only gaps, which is economically obvious and here algebraically automatic.

Positive weights make the graph connected, so the constant vector is the *only* null direction, the gradient map is strictly monotone as an operator on $\mathbf{1}^\perp$ (not coordinatewise), and shares determine utilities uniquely.

And symmetry plus positive definiteness make every reduced Newton system SPD, so conjugate gradients apply with the $O(QNL)$ Jacobian–vector product of Proposition 5 playing the role of a fast Laplacian multiply. Structurally, each linearization of the calibration solves a weighted graph-Laplacian system on the photo-finish graph (the weights themselves move with μ), and such systems are easy for the same reason resistor networks and heat equations are.

None of this is specifically Gaussian: any conditionally independent additive random-utility race with everywhere-positive smooth densities has, under the corresponding integrability and differentiability conditions (finite social surplus, differentiation under the integral), a weighted-Laplacian Jacobian and an interior-simplex diffeomorphism modulo translation; Gaussian factor probit is the computationally important specialization.

Remark 2. Social-surplus convex duality for discrete choice is classical (Chiong et al., 2016; Norets and Takahashi, 2013); what is new here is not well-posedness but its scalable realization. The bijection itself is essentially known: injectivity up to a common shift is the Hotz–Miller inversion (Hotz and Miller, 1993), surjectivity is Norets and Takahashi (2013), and Berry et al. (2013) derive demand invertibility from connectivity of the substitution structure, the same mechanism as the photo-finish graph; the compact proof above keeps the Gaussian factor case self-contained and exhibits the weights the algorithm uses. Each identity in the proposition is also checked numerically in the repository: the w_{ij} representation against finite differences to 6×10^{-9}, symmetry to 4×10^{-12}, and positive definiteness of the reduced numerical Jacobian.

Proposition 5 (Jacobian–vector products in $O(QNL)$). *With hazards $\lambda_j = g_j/F_j$, $\Lambda = \sum_j \lambda_j$, and $A = \sum_j h_j\lambda_j$ (all conditional on f, all functions of x), the Jacobian of the exact max-wins map acts on a direction h as*

$$(Jh)_i \;=\; \mathbb{E}_f \int g_i\Big(\prod_{j\neq i} F_j\Big)\,(h_i\Lambda - A)\,dx,$$

where the field sums Λ and A are formed once per node and lattice point, so all N components cost $O(NL)$ per node. In English: a direction h perturbs alternative i's share only through the gap between its own hazard weight $h_i\Lambda$ and the field average A. Translation invariance is visible ($h = \mathbf{1}$ gives $A = \Lambda$, hence $Jh = 0$), and the weighted-Laplacian form of Proposition 4 follows by expanding A. This enables Krylov solves of the reduced system in Proposition 3 without forming J densely at $O(QN^2L)$.

Proof. Write $R_i = \prod_{j\neq i} F_j$. For the own derivative, $\partial_{\mu_i} g_i(x) = -g_i'(x)$ (a location shift), and integrating by parts, $\int(-g_i')R_i\,dx = \int g_i R_i'\,dx = \int g_i R_i \sum_{j\neq i}\lambda_j\,dx = \int g_i R_i(\Lambda - \lambda_i)\,dx$. For $j \neq i$, $\partial_{\mu_j} F_j = -g_j$, so the cross derivative is $-\int g_i g_j \prod_{\ell\neq i,j} F_\ell\,dx = -\int g_i R_i \lambda_j\,dx$. Summing against h, $(Jh)_i = \int g_i R_i[\,h_i(\Lambda - \lambda_i) - \sum_{j\neq i} h_j\lambda_j\,]dx = \int g_i R_i(h_i\Lambda - A)\,dx$; average over f. $\square$

The differentiated map. Three maps must be distinguished: the exact max-wins map $p^{\max}(\mu)$; the reflected min-wins map $p^{\min}(a)$ used by the implementation, related by $a = -\mu$ so that $D_\mu p^{\max}[h] = D_a p^{\min}[-h]$; and the implemented approximation $p_{Q,L} = \tilde{p}_{Q,L}/(\mathbf{1}^\top \tilde{p}_{Q,L})$, whose derivative carries a normalization correction: with $v = D\tilde{p}[h]$ and $s = \mathbf{1}^\top\tilde{p}$,

$$Dp_{Q,L}[h] \;=\; \frac{v - p_{Q,L}\,(\mathbf{1}^\top v)}{s}.$$

The correction vanishes for the continuum map and is numerically negligible at the production resolutions tested. A further distinction: integration by parts is a continuum identity, so the Laplacian formula of Proposition 5 is the exact derivative of the *continuum* map, not of the finite rectangle sum.

The exact derivative of the frozen-grid sum replaces the own-term by $(x - m_i)/D_i$ (no integration by parts), costs the same $O(QNL)$, and ships in the repository (`form="grid"`).

Measured at production resolution ($L = 1501$) the two forms agree to 1.7×10^{-14}; at $L = 51$ the continuum form is off by 2.6×10^{-5} while the grid form still matches finite differences.

So the Laplacian form is an extremely accurate symmetric surrogate at the resolutions used, and exact differentiation of the normalized *frozen-grid* map is available when needed; the returned adaptive map additionally moves its envelope with μ, a tail effect measured below 10^{-8} at production resolution.

The implemented inversion rebuilds the lattice envelope from the current utilities at each iteration, holds it fixed within the iteration, and projects the utilities to mean zero after every update (Algorithm 2).

Tail-stable implementation. Tail stability requires the log domain throughout: $\log \Phi$ evaluated directly (forming $1-\Phi(z)$ by literal subtraction rounds to zero near $z \approx 8.3$ through cancellation), the analytic $\log g$, and hazards as $\exp(\log g - \log \Phi)$; the Gaussian inverse Mills ratio is unbounded but grows only linearly in the extreme tail, rather than exponentially, which is the numerically important property.

Flooring a density before taking its logarithm while the exact log-survival stays finite produces spurious e^{+92} hazards on problems as mild as $N = 8$ with heterogeneous D; that configuration is a regression test. In the forward map the same discipline lets deep-tail shares resolve to genuine small positives rather than floored zeros.

Beyond Gaussian factors. The factors need not be Gaussian: (2) requires only conditional independence and an integrable factor law, so Student, mixture, or empirical factor distributions are admissible with appropriate nodes, and the base densities may be arbitrary and alternative-specific. One caution for asymmetric laws: the implementation works in the reflected (min-wins) race, and $-U$ has the same distribution as the reflected model only when the factor and base laws are symmetric; for asymmetric laws the reflected distributions must be used explicitly.

Numerical method. All experiments use $L = 501$ lattice points, chosen from the measured lattice sweep (experiment 17 spans $L = 101$ to 6001; self-convergence reaches machine precision by $L \approx 200$), on the adaptive interval $[\min_{i,q} m_i(f_q) - 8 \max_i \sqrt{D_i},\ \max_{i,q} m_i(f_q) + 8 \max_i \sqrt{D_i}]$, where $q = 1, \dots, Q$ indexes the retained factor nodes.

Factor integration: the main experiments use product Gauss–Hermite of order 15/11/9 for $k = 2/3/4$; the boundary experiment (experiment 14) uses order 15 per dimension for all $k \le 4$, which after pruning weights below 10^{-7} of the maximum leaves $Q = 145$, 1317, and 10929 nodes for $k = 2, 3, 4$, making the exponential growth of Q with rank explicit. Scrambled-Sobol nodes (2^{13} points, fixed seed) are used for $k > 4$: fixed-node and reproducible (a fixed-seed randomized Sobol construction), rather than deterministic in the strict sense.

Survival floors sit at 10^{-300} and the quadrature output is normalized by its sum. The returned map is a reproducible, piecewise-smooth numerical approximation of (2); statements about derivatives refer to derivatives of this approximation, whose fidelity to exact MNP derivatives is governed by L and Q.

The rectangle rule uses Δx weights with no endpoint modification (differing from the trapezoidal rule only by negligible endpoint terms).

Pruned factor weights are not renormalized explicitly; the final normalization conditions the quadrature on the retained node set, and the two contributions to the pre-normalization defect separate cleanly: the pruned factor weight is $\delta_{\text{factor}} = 2.39 \times 10^{-8}$ at $k = 2$ (analytic), and the measured $|1 - \mathbf{1}^\top \tilde{p}| = 2.4 \times 10^{-8}$ equals it, so lattice truncation contributes below that level.

Including the conditional-location build, the complete forward cost is $O(QN(k + L))$, reducing to $O(QNL)$ for $k \ll L$.

The slopes in Algorithm 2 ignore the moving envelope and the output normalization;

Algorithm 1 All-share forward pass (min-wins; reflected from argmax by $a = -\mu$)

Require: abilities a, loadings V, idiosyncratic variances D, factor nodes $\{f_q, w_q\}_{q=1}^{Q}$, lattice size L

$m_{iq} \leftarrow a_i + v_i^\top f_q$ for all i, q

$[x_1, x_L] \leftarrow [\min_{i,q} m_{iq} - 8 \max_i \sqrt{D_i},\ \max_{i,q} m_{iq} + 8 \max_i \sqrt{D_i}]$, uniform spacing Δx

for $q = 1, \ldots, Q$ **do**

  $\log S_i(x_\ell) \leftarrow \log \Phi\big(-(x_\ell - m_{iq})/\sqrt{D_i}\big)$ for all i, ℓ ▷ never $1 - \Phi$

  $\log S_{\text{field}}(x_\ell) \leftarrow \sum_j \log S_j(x_\ell)$ ▷ one shared field

  $\tilde{p}_i \mathrel{+}= w_q \sum_\ell g_i(x_\ell) \exp\{\log S_{\text{field}}(x_\ell) - \log S_i(x_\ell)\} \Delta x$ ▷ divide one out

end for

return $p = \tilde{p}/(\mathbf{1}^\top \tilde{p})$ and the diagnostic $|1 - \mathbf{1}^\top \tilde{p}|$

Algorithm 2 Calibration (damped Jacobi quasi-Newton)

Require: strictly positive target shares $\hat{p}$ (normalized), V, D, factor rule, tolerance 10^{-6}, cap 50 iterations

$a \leftarrow$ exact independent inverse using total variances $D_i + \|v_i\|^2$ ▷ warm start

repeat

  run Algorithm 1 at a, collecting $\tilde{p}$ and the analytic own-location slopes $\partial_i = \partial \tilde{p}_i / \partial a_i$ from the same pass

  $r_i \leftarrow \log(\tilde{p}_i / \mathbf{1}^\top \tilde{p}) - \log \hat{p}_i$; residual $\leftarrow \max\{|r_i| : \hat{p}_i > 10^{-4}/N\}$

  halve the damping (floor 0.25) if the residual grew by more than 20%

  $a_i \leftarrow a_i - \text{damp} \cdot r_i / (\partial_i / \tilde{p}_i)$, steps capped at $\sqrt{D_i + \|v_i\|^2}$

  $a \leftarrow a - \bar{a}$ ▷ project to mean zero every update

until residual $< 10^{-6}$

they are Jacobi quasi-Newton preconditioners, not the exact diagonal of the adaptive map's Jacobian.

Inversion. Given target shares $\hat{p}$ and (V, D), the inverse transform finds mean-zero μ with $p(\mu) = \hat{p}$ by Algorithm 2. The tolerance is tail-aware (alternatives with target shares below $10^{-4}/N$ are matched best-effort; by Proposition 4 they remain uniquely determined, but they are statistically weakly resolved, which is also a conditioning caution for the reduced Jacobian when shares are tiny or alternatives nearly collinear).

Strictly zero observed shares lie outside the proposition: boundary targets correspond to utilities diverging to $-\infty$, so real assortment data requires pseudocounts or regularization, not merely the implementation floor.

Convergence in a handful of iterations is an empirical observation on the tested problems, not a theorem. The design follows the original transform's inversion (Cotton, 2021) and the calibration practice of Cotton (2024).

2 Validation

Correctness anchors. On the binary case, where MNP has a closed form, the lattice transform agrees to 3.3×10^{-16} (an easy anchor, useful for code validation rather than as evidence about the hard regime).

At $N = 5$ both methods sit within the noise of a 10^7-draw Monte Carlo reference, and the shipped `thurstone.FactorRace` agrees with the benchmark kernel to 1.1×10^{-5}.

Error metric. Throughout, $\max_i |\hat{p}_i - p_i|$ against a simulation reference. Absolute error is not always the decision-relevant scale: in betting, moving a 100-to-1 chance to 150-to-1 is an enormous change carried by $|\Delta p| \approx 0.003$, so log-odds error matters, and there the log-domain lattice is at its strongest (deep-tail shares resolve to genuine small positives) while a simulation reference certifies relative error ϵ only for shares above roughly $1/(R\epsilon^2)$; committed records therefore carry both metrics. A maximum over N coordinates tends to grow with N at fixed per-coordinate accuracy, and the reference's own maximum-coordinate noise behaves likewise; mean-absolute and total-variation summaries are committed alongside (experiment 16): at $N = 1000$ the lattice's mean absolute error is 5.6×10^{-6} and total variation 2.8×10^{-3} against a fresh 2×10^6-draw reference.

Replication. Ten problems per size at $N \leq 200$, three at $N = 1000$, each scored against twin independent 2×10^6-draw references (experiment 16): worst-case lattice error 3.5–4.6×10^{-4} at every size, at or below the references' own replicate-to-replicate noise. The lattice's true accuracy is far below this measurement floor, as the resolution sweeps and log-odds certification show.

Log-odds accuracy. Against a super-converged reference (Gauss–Hermite 60^2, $L = 5001$; experiment 25) the production settings' log-odds error is at most 5×10^{-5} on every share from 10^{-9} to one. No simulation reference could certify this: pinning relative error ϵ on a share p takes about $1/(p\epsilon^2)$ draws.

Resolution sweeps. Experiment 17 sweeps each resolution knob separately at $N = 200$, $k = 2$. Holding the factor rule fixed, lattice self-convergence is 2.7×10^{-8} at $L = 101$ and at machine precision from $L \approx 200$; the convergence observed over the tested range is near-exponential, and for these analytic, rapidly decaying integrands the lattice error becomes negligible very quickly, leaving the L-error subdominant.

Holding L fixed, the Gauss–Hermite sweep decays from 5.3×10^{-4} (order 3) to 5.9×10^{-9} (order 15).

The pre-normalization defect $|1 - \mathbf{1}^\top \tilde{p}\,|$ is 2.4×10^{-8}.

Derivatives. For fixed factor nodes and a fixed lattice, the transform is a reproducible, (piecewise-)differentiable function of μ whose derivatives are evaluated without resampling; analytic own-location slopes of the unnormalized map come from the same pass and serve as the inversion preconditioner. Along a utility path at $N = 50$, the maximum second difference over dt^2 is 39.3 for fresh-draw GHK against 0.01 for both CRN-GHK and the lattice; the committed smoothness figure shows the curves.

These are derivatives of the numerical approximation, not exact MNP derivatives, and GHK-based practice likewise differentiates its approximation, analytically or with fixed draws (Hajivassiliou and McFadden, 1998; Train, 2009). The practical distinction is the character and controllability of the approximation error, quadrature resolution versus draw count, not the possibility of differentiation.

3 Calibration at $N = 1000$

At $N = 1000$, $k = 2$, with targets from 5×10^6 MC draws floored at 10^{-7} and renormalized: the inversion completes in 3 seconds compiled, 19 in the reference NumPy implementation (all subsequent unlabeled timings are NumPy). The iteration solves its own equations to 6.4×10^{-9} (a statement about the solver, not statistical accuracy, since the target carries sampling noise near 2×10^{-4}) and recovers the utilities of the well-resolved alternatives (target share above 3×10^{-4}) to maximum error 0.017.

A replication study (experiment 16) repeats the inversion on a second $N = 1000$ problem against three independent 5×10^6-draw targets: 18 NumPy seconds each, 161–162 alternatives well-resolved (98.1% of share mass), recovery error maximum 0.015–0.023 and median ≈ 0.0027 across replicates, with the recovery-versus-share curve committed alongside.

Validation: where the error comes from. Experiment 21 separates the error sources. A noiseless inverse test (targets generated by the lattice map itself, a deliberate inverse crime that isolates solver plumbing) recovers the utilities of every resolvable alternative to 6.5×10^{-7} at $N = 200$ and 1.2×10^{-7} at $N = 1000$ — solver error at the tolerance, so the ≈ 0.02 recovery in noisy tests is target sampling noise, not solver error. Alternatives with shares below the resolution floor are not chased, by design; their tier is reported separately.

Robustness. Twenty problems that vary utility spread, factor strength, D-heterogeneity, and rank all converged (6–31 iterations, recovery 0.004–0.024).

Numerical well-posedness matches Proposition 4: the reduced Jacobian assembled from N JVP calls at $N = 50$ is symmetric to 1.7×10^{-18}, has row-sum defect 6.2×10^{-17}, and reduced eigenvalues in $[2.5 \times 10^{-5}, 0.22]$, positive definite as proved.

Stability checks and two genuine bounds. Self-convergence checks at large N show stability below what the simulation references can resolve: refining to $(L = 3001, \mathrm{GH21})$ moves the shares by 1.2×10^{-8} at $N = 1000$ and 5.7×10^{-8} at $N = 5000$ (a stability statement, not by itself an error bound, since both resolutions could share bias).

Two genuine bounds come free from the product structure: conditional probabilities lie in $[0, 1]$, so pruning factor weight δ perturbs unnormalized shares by at most $\delta =$

2.39×10^{-8} in sup norm (measured: 6.2×10^{-9}); and the $\pm 8\sigma_{\max}$ envelope omits conditional mass at most $N\Phi(-8) \approx 6 \times 10^{-12}$ even at $N = 10^4$.

Compiled implementation. The same pass in Rust (in the repository, with machine-precision parity and identical iteration counts) calibrates $N = 1000$ in 3 seconds, $N = 5000$ in 18 seconds, and $N = 10000$ in 38 seconds (10, 11, and 13 iterations; independent 5×10^6-draw targets, experiment 19).

Scaling. Approximately linear in N over the tested range (experiment 19; at fixed spatial resolution the lattice would grow with the utility range, $O(\sqrt{\log N})$ under Gaussian utilities): NumPy calibration takes 16, 41, 110, and 201 seconds at $N = 1000, 2000, 5000, 10000$, a flat 11–15 forward-pass equivalents throughout, with recovery quality unchanged (median error near 0.003, 96–99% of share mass identified).

Why simulation cannot follow. The gap here is structural, and simple arithmetic shows why (a plain independent Monte Carlo sample-complexity calculation, not a lower bound on simulation methods generally). With common random numbers a smoothed simulator such as CRN-GHK is a fixed function that a solver can match to fine tolerance (a raw draw-and-argmax map is piecewise constant on the $1/R$ grid and cannot be); either way the limitation is that it is the wrong map.

Its coordinatewise error against the exact map is of order $\sqrt{p_{\max}/R}$, so making the simulated map itself τ-accurate costs $R \approx p_{\max}/\tau^2$ independent draws per evaluation, and inverting it exactly still leaves utilities whose true shares miss the target at scale τ, amplified through J^{-1}.

Under plain independent direct simulation at the measured $N = 5000$ throughput, this prices a forward map resolving the *exact* MNP map to $\tau \sim 10^{-6}$ (this solver's model-consistency level, well beyond the 2×10^{-4} statistical accuracy of the noisy-target experiments) at weeks per evaluation and the calibration in months to a year, against seconds here: roughly a million times slower at equivalent accuracy.

Common random numbers do not change the plain-MC rate; QMC and specialized Gaussian integration improve the constants without changing the structure, and a maximum over N coordinates adds a further $\log N$ factor in high-probability statements.

At the crude $\tau = 10^{-4}$ where simulation-in-the-loop first becomes feasible, the calibrated model's true shares miss the target at 10^{-4} and the answer moves with the draw set.

The quadrature route's accuracy against the exact map is set by resolution, not draws, at cost independent of τ until resolution binds. We are unaware of a reported probit share inversion at this scale in simulation-based practice, and this scaling is presumably why.

Removal ensemble (by-product). The cached field also yields all N single-removal share vectors in $O(QN^2L)$: 15.9 seconds for all 200 removals at $N = 200$, against 56.8 seconds recomputing.

Against the strongest simulation comparator, which reuses each draw's winner and runner-up to answer every removal at once in $O(RN + N^2)$, the deterministic ensemble is about $4.5\times$ more accurate at matched wall time at $N = 200$ (4.2×10^{-5} versus 1.9×10^{-4}; experiment 18). This is a side capability, not the paper's case.

4 Boundary experiments

General covariance through the factor approximation. The method requires $\Sigma \approx VV^\top + \mathrm{diag}(D)$; GHK does not. Choices depend on (μ, Σ) only through $P\mu$ and $P\Sigma P$, $P = I - \mathbf{1}\mathbf{1}^\top/N$, because the argmax is determined by pairwise differences: a common factor loading $V \to V + \mathbf{1}c^\top$ shifts every conditional location equally and cannot move the argmax (a machine-precision identity), so the factor fit must be run in this choice-relevant quotient space.

Fitting raw Σ can spend scarce rank on an irrelevant common component ($\Sigma = \tau^2\mathbf{1}\mathbf{1}^\top + \beta\beta^\top + D$ with large τ: the raw rank-1 fit takes $\mathbf{1}\mathbf{1}^\top$ and misses $\beta\beta^\top$ entirely; on such an example the contrast-space fit cuts rank-1 share error from 4.6×10^{-2} to 8.3×10^{-3}).

The asymmetry matters: only the common *factor* direction is irrelevant; a common addition to D inflates every difference variance and is choice-relevant, so D must come from the quotient fit.

The fit itself is an iterative principal-factor heuristic applied to $P\Sigma P$ (eigendecompose $P\Sigma P - \mathrm{diag}(D)$, take the top k directions, refit D from the diagonal with a 10^{-3} floor, iterate to 10^{-10}), followed by centering and SVD canonicalization of the loadings (ordered columns, fixed signs).

It is not certified to minimize a projected norm ($P\,\mathrm{diag}(D)\,P$ is not diagonal), so the repository also implements an alternating least-squares fit with exact block updates on the reduced quotient (top-k PSD step for the loadings, nonnegative least squares for D; exact block minimization, not a global-optimality certificate): on the boundary matrices it changes the quotient residual by under 1%, so the heuristic is not the binding constraint.

The covariance-quotient residual $\|P(\Sigma - \hat{\Sigma})P\|_F / \|P\Sigma P\|_F$ is reported alongside every share error in the committed output.

With this fit: correlation matrices at spectral decay $\lambda_m \propto m^{-\gamma}$, $\gamma \in \{0.5, 1.5, 3\}$, at $N = 50$, sharing one orthogonal eigenbasis across γ so decay comparisons are not confounded with basis realization. The power law holds before diagonal standardization, which perturbs the spectrum; the actual leading eigenvalues (3.7, 17.1, 34.2 at $\gamma = 0.5, 1.5, 3$) are printed by the script.

Share error is measured against an 8×10^6-draw reference, with GHK at the exact Σ for comparison; the plotted quantity is the rank-k approximation error of the stated procedure, not a minimum over factor models.

Rank-8 errors: 1.8×10^{-3}, 3.7×10^{-3}, and 3.1×10^{-3} at $\gamma = 0.5, 1.5, 3$ from $k = 1$ starting points of 2.3×10^{-3}, 1.5×10^{-2}, 7.1×10^{-2}.

Decomposing the error with a second simulation under the fitted factor model: integration error is flat at $2\text{–}7 \times 10^{-4}$ across every (γ, k) tested, so the rank-k error is essentially covariance-fit error: rank, not quadrature, is the binding constraint.

Against GHK at $R = 10^4$ (3.2, 2.3, 4.3×10^{-3}), rank-8 wins at $\gamma = 0.5$ and 3 but *loses at intermediate decay* $\gamma = 1.5$ (3.7×10^{-3} versus 2.3×10^{-3}): for this construction the intermediate case is empirically hardest. Diagonal standardization perturbs the nominal power law and the leading eigenvalue varies from 3.7 to 34.2 across the three cases, so this is not a clean one-parameter family.

Replication over three independent eigenbases: rank-8 beats GHK at $R = 10^4$ in 8 of 9 basis–decay cases, with the single loss at intermediate decay. The intermediate construction is empirically hardest, but this is not a stable GHK-wins regime.

Nor is this wall-time-matched: the $k = 8$ RQMC evaluation takes 15–24 s against 0.5–0.9 s for GHK at $R = 10^4$, so at $N = 50$ GHK is the cheaper route to $3\text{–}4 \times 10^{-3}$; the lattice's advantage is large N at small k. Where accuracy demands exceed the achievable rank-k error, GHK's arbitrary-Σ generality is the right tool.

Substitution: a 2×2 factorial illustration. The design varies factor structure and noise family one axis at a time, holding the idiosyncratic scale common. Coupling either axis with alternative-specific scales would confound the comparison: an alternative-specific-scale Gumbel race is not mixed logit, and heteroskedasticity alone already breaks IIA. A separate heteroskedasticity axis is left to future replication.

Ground truth ($N = 50$) is misspecified for every candidate: $t(5)$ factors (unit variance) and skew-normal idiosyncratic noise (shape $a = 3$), standardized to mean zero and unit variance, homoskedastic, with $\mu^\star \sim N(0, 1)$ and oracle loadings $v^\star_{ir} \sim N(0, 0.36/k)$, $k = 2$. (The race is computed min-wins, so positive skew in race coordinates is *left*-skew in utility coordinates.)

The four candidates hold the idiosyncratic scale fixed and vary $\{\text{Gumbel}, \text{Gaussian}\}$ base $\times$ $\{V = 0, V = V^\star\}$. The Gumbel base is variance-standardized to one, so the family axis does not alter the factor-to-idiosyncratic variance ratio.

Because the implementation is min-wins, the Gumbel base is the *reflected* (minimum) Gumbel corresponding to a standard maximum-Gumbel in utility coordinates; the committed anchor verifies that its zero-loading case equals softmax at $N = 12$ to 2.8×10^{-17}. The candidate factor law is Gaussian in all cases, so the truth's $t(5)$ factor law is misspecified for every candidate.

This is an *oracle-loading* design: both factor candidates receive the true $V^\star$ rather than estimating it. All four are calibrated to the same full-menu shares (residuals $\leq 4 \times 10^{-10}$).

Deletion truths come from 2×10^7 *common* utility draws with the top three alternatives retained per draw, from which every single-deletion and pair-deletion truth follows with common random numbers.

Of 150 deletion blocks (50 singles, 100 fixed-seed random pairs), 45 with deleted mass below 0.05% are excluded as unresolvable; of the 105 scored blocks, 83 fall in the three best-resolved strata (26/29/28 at >10% / 2–10% / 0.5–2%) and the remaining 22, with deleted mass between 0.05% and 0.5%, appear in the final column (their truths sit closest to the common-draw resolution floor, so that column carries the widest error bars).

The score per deletion block B is the misallocated fraction of the redistributed mass,

$$\mathrm{TV}\Big(q^{(-B)}_{\text{model}},\, q^{(-B)}_{\text{truth}}\Big) \Big/ \textstyle\sum_{i \in B} p_i,$$

averaged within strata:

	mass > 10%	2–10%	0.5–2%	0.05–0.5%
independent Luce (= plain logit)	0.229	0.245	0.261	0.305
independent probit	0.210	0.232	0.230	0.267
factor mixed logit	0.114	0.146	0.199	0.244
factor probit	**0.045**	**0.062**	**0.094**	**0.116**

The factorial decomposition (top two strata): the factor increment is dominant in both families ($+0.115/+0.099$ within Gumbel, $+0.165/+0.170$ within Gaussian); the family increment alone is small ($+0.019/+0.013$ at $V=0$) but grows with factors present ($+0.069/+0.084$), a negative interaction ($-0.050/-0.071$): factor structure and the Gaussian base are complements in this design.

Caveats: a single truth design and seed, and a skew-normal truth that is Gaussian-adjacent. The figure plots, for each scored single deletion, plain logit's misallocation against factor probit's: points below the diagonal are deletions where factor probit is closer to the truth. Singles and pairs score similarly (factor probit 0.080 versus 0.076 mean misallocated fraction) and are summarized separately in the committed output.

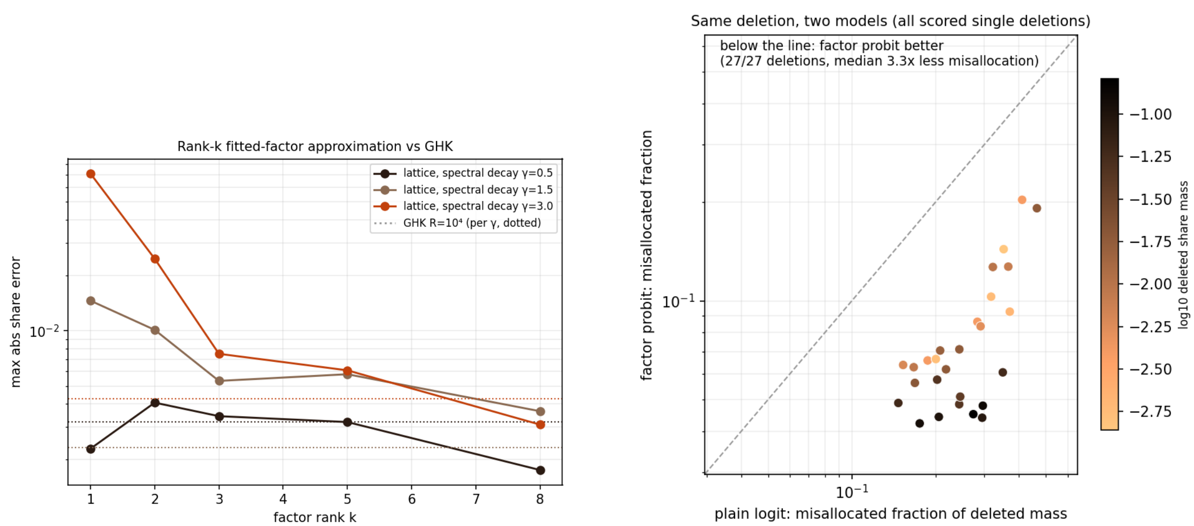

Figure 2: Left: rank-k contrast-space approximation error under spectral decay γ (dotted: GHK at $R=10^4$, per γ; not wall-time matched). Right: paired comparison on identical deletions — plain logit's misallocated fraction (horizontal) against factor probit's (vertical), one point per scored single deletion, shaded by deleted mass; the dashed line is parity.

5 Related work

GHK and simulated likelihood are standard in econometrics (Geweke et al., 1994; Hajivassiliou and McFadden, 1998; Train, 2009); Genz (1992) developed the same separation-of-variables construction in numerical statistics, where its RQMC form is the default MVN evaluator. Ridgway (2016) reads GHK as sequential importance sampling, proves its variance can deteriorate exponentially, and builds SMC alternatives; Botev (2017) sharpened the family with minimax tilting; Cao et al. (2021) push MVN probabilities to thousands of dimensions with tile-low-rank QMC, machinery pointed almost exactly at our difference covariances, which are diagonal plus rank $k + 1$.

Deterministic approximations run from Mendell–Elston (Mendell and Elston, 1974) (surprisingly competitive when optimally ordered (Connors et al., 2014)) through Bhat's MACML (Bhat, 2011) to expectation propagation: Cunningham et al. (2011) found EP highly accurate precisely for rectangular regions, and Fasano and Denti (2025) evaluate generic Gaussian CDFs by EP on a dual probit model, deterministically and sampling-free. Every one of these prices one orthant at a time; none couples the N winner probabilities.

Factor dimension reduction for probit is old, and older than econometrics: Curnow and Dunnett (1962) showed that product correlation collapses the multivariate normal integral to a one-dimensional integral of products of univariate CDFs, the reduction this paper conditions on, and Dunnett (1989) is its standard algorithm. The ranking-and-selection tradition built on the same integrals from the 1950s (Bechhofer, 1954; Gupta, 1963), always evaluating one designated candidate per integral: assembling all N that way costs N separate integrals where the shared field gives all N in one pass. In econometrics, Butler and Moffitt (1982) integrated one factor by quadrature; Elrod and Keane (1995) estimated factor-analytic probit for panel data; Loaiza-Maya and Nibbering (2022) pursue large choice sets by Bayesian estimation, with variational follow-ups (Loaiza-Maya and Nibbering, 2023); Kim et al. (2026) reach twenty alternatives at a million observations by neural variational inference, and Ding et al. (2024) exceed a hundred with expectation-propagation E-steps. All are approximate forward maps for estimation; none inverts shares. Girolami and Rogers (2006) use the one-dimensional product identity in its diagonal-covariance form, per class. The racing literature posed both ingredients first: Henery (1981) wrote the survival-product win integral for normal running times and gave its first-order analytic inversion, Ali (1998) inverted market odds to abilities numerically at eight-runner fields, and betting practice otherwise approximated the inversion away (Lo and Bacon-Shone, 1994). The lattice survival-product identity and its fast inverse transform come from racing (Cotton, 2021), which names the common-factor extension developed here as an open problem.

The shared-product assembly itself also appears in competing-risks computation: Mahani and Sharabiani (2019) compute all cause-specific cumulative incidences on one shared grid by forming the product of every survival function once and dividing out each cause's own factor (software since 2015), and the Aalen–Johansen estimator (Aalen and Johansen, 1978) reads all causes off a single survival product nonparametrically. Neither embeds the assembly in factor quadrature, differentiates it, nor inverts it; racing arrived at it independently.

Conditioning on the factors and the alternative's own shock makes each probability a $(k+1)$-dimensional integral,

$$p_i = \mathbb{E}_{f,z} \prod_{j \neq i} \Phi\left(\frac{\mu_i - \mu_j + (v_i - v_j)^\top f + \sqrt{D_i}\, z}{\sqrt{D_j}}\right),$$

evaluable per alternative by modern randomized quasi-Monte Carlo: the natural competitor that uses exactly the structure this paper assumes. It still prices N separate integrals, $O(RN^2)$ in total, where the shared field couples them into one $O(QN(k+L))$ pass; experiment 24 measures the gap.

Inversion theory is likewise established. Hotz and Miller (1993) inverted choice probabilities to utility differences; Norets and Takahashi (2013) prove surjectivity of the utility-to-probability map; Berry et al. (2013) obtain demand invertibility from connected substitution; Chiong et al. (2016) invert through the convex conjugate; Fosgerau et al. (2020) show the probit inverse exists by generalized-entropy duality, without an algorithm; Berry et al. (1995) built demand estimation on the logit-side inversion, and Conlon and Gortmaker (2020) maintain its modern practice. Chia (2021) differentiates logit BLP automatically; Huch and Keane (2026) amortize correlated-choice probabilities with neural networks, evidence of current demand for this computation.

The novelty here is accordingly none of: dimension reduction (Curnow and Dunnett, 1962), the order-statistic integral, the shared-product assembly in isolation (Mahani and Sharabiani, 2019; Cotton, 2021), or well-posedness of inversion. It is the all-alternatives assembly in $O(QNL)$, its large-N implementation, the $O(QNL)$ Jacobian–vector product, and the shared-field reuse for calibration and counterfactuals.

6 Discussion

Estimation is the natural next step. With the reduced coordinates of Proposition 3, gradients of an outer likelihood or GMM objective pass through the calibration fixed point by implicit differentiation, a route to differentiable probit demand estimation at large N. Estimating (V, D) rather than supplying them raises the usual identification cautions: scale normalization, factor rotation, and the fact that only difference covariances are identified.

The pass itself can be accelerated further: its two stages are smooth-kernel sums whose matrices are numerically low-rank, and a Chebyshev-separated evaluation (in the spirit of black-box fast multipole methods) reduces each node's cost from $O(NL)$ to $O(r(N+L))$; a committed prototype (experiment 20) reproduces the exact pass to 6×10^{-5} at $23\times$ speed for $N = 5000$ in NumPy, and the compiled version of the same construction measures $93\times$ (77 ms for all 5000 shares), with exponential convergence in r. Development to full generality is left to a companion note.

Beyond the Gaussian family, the same machinery computes races with arbitrary alternative-specific idiosyncratic densities. A common-scale Gumbel base with zero loadings reproduces the Luce model exactly, yielding correlated-softmax generalizations.

Neither method class dominates the other. Simulation owns arbitrary Σ and coarse accuracy; the shared field owns the factor form, tight accuracy, derivatives, and calibration; the boundary experiments draw the border at intermediate spectral decay. We leave to future work replication of the substitution factorial across truth designs and a walk-forward calibration on real assortment data.

Supplementary Materials

Code and experiments: all numbers come from committed, seeded scripts with correctness anchors run before any comparison (repository experiments 13–25). A single manifest, `experiments/run_all_paper.py`, regenerates every table and figure; the exact commit is recorded in the repository README and an immutable tag is pinned at submission. (Archive supplied as a zip with the submission; development repository at `https://github.com/microprediction/kinetics`.)

Software: the algorithm ships in an open-source Python package with an optional compiled (Rust) kernel, and package–benchmark agreement is itself a reported number.

Data: all experiments use synthetic data generated by the seeded scripts; no external data are required.

Hardware and software: Apple M4, 16 GB, Python 3.12.9, NumPy 2.4.6, SciPy 1.18.0; experiment 17 enforces single-threaded BLAS via `threadpoolctl`; sub-second timings are warmed medians of three runs, and long simulations are single realizations, disclosed as such. Monte Carlo chunk sizes derive from an explicit 1.5 GB memory budget, and peak resident memory stays below 2 GB.

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